12 / LTSPICE

Circuit Simulation & Correlation

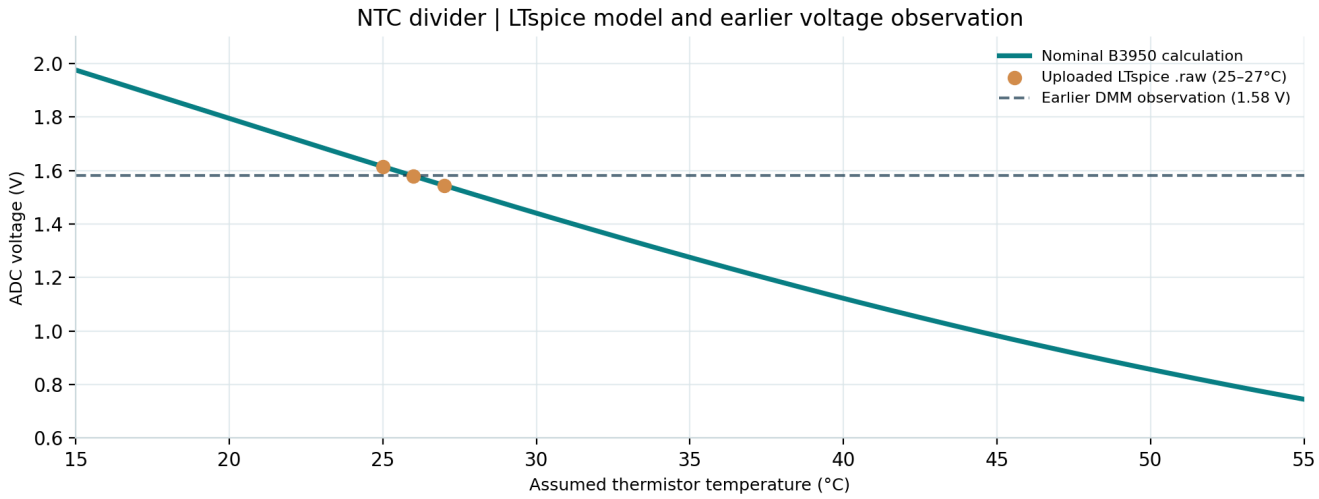
User-run .cir, .log and .raw files included with this project

LTSPICE TEMP STEPS	AC SWEEP POINTS	CUTOFF (MODEL)
3	501	267.7 Hz
25°C, 26°C, 27°C	1 Hz through 100 kHz	25°C nominal NTC model

01 / MODEL AND SIMULATION SCOPE

The uploaded LTspice source models a 3.2 V supply, R1 = 9.82 kΩ, a nominal 10 kΩ B3950 NTC sensor, R2 = 990 Ω and C1 = 100 nF. GPIO34 is represented as a high-impedance analog input, not a complete ESP32 ADC electrical model.

02 / TEMPERATURE-DEPENDENT VOLTAGE DIVIDER



Model-derived temperatures are assumptions in the simulation. An independent reference thermometer was not recorded with the earlier 1.58 V breadboard observation; no temperature calibration is claimed.

03 / SOURCE EVIDENCE

LTspice .cir files preserve the model and operating point; original .raw files preserve waveform/frequency data; original .log files preserve solver and step results. Graph datasets were extracted directly from the uploaded .raw files.

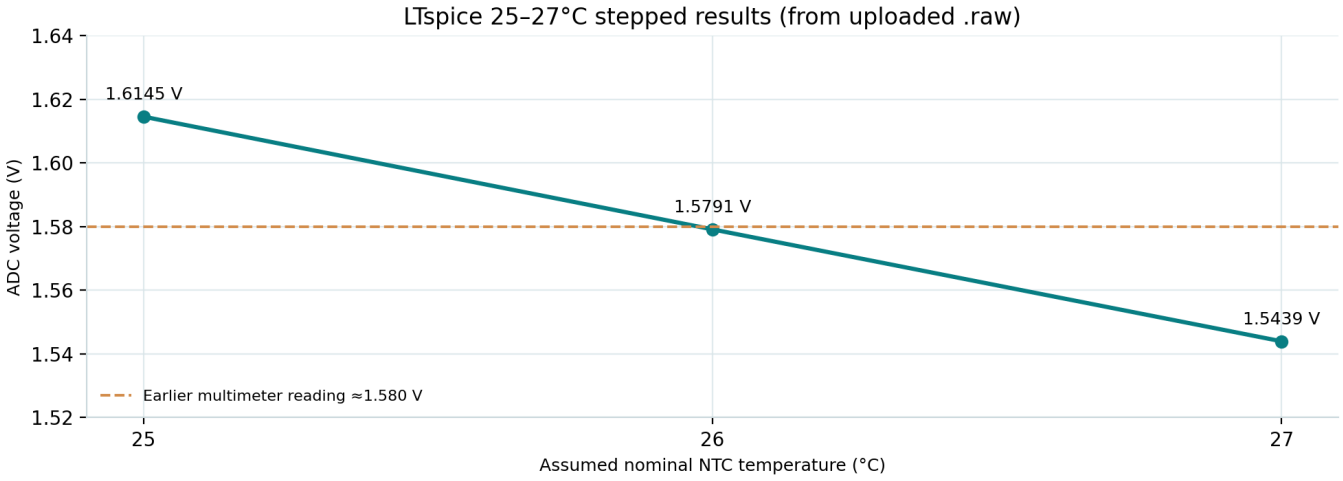
13 / TEMPERATURE

Voltage Predictions vs Breadboard

Real multimeter observation compared with nominal LTspice steps

01 / VERIFIED LTSPICE STEP VALUES

ASSUMED NTC TEMP	LTSPICE ADC	ORIGIN
25°C	1.614531 V	Actual .raw / .log
26°C	1.579103 V	Actual .raw / .log
27°C	1.543931 V	Actual .raw / .log



02 / MEASURED-VERSUS-MODELED OBSERVATION

An earlier breadboard measurement on the actual thermistor was about 1.580 V at the ADC input (3.20 V supply). A nominal 26°C model produces 1.579103 V, a difference of about 0.90 mV. This means the observed voltage is consistent with the model if the thermistor was near 26°C; it does not establish its actual temperature.

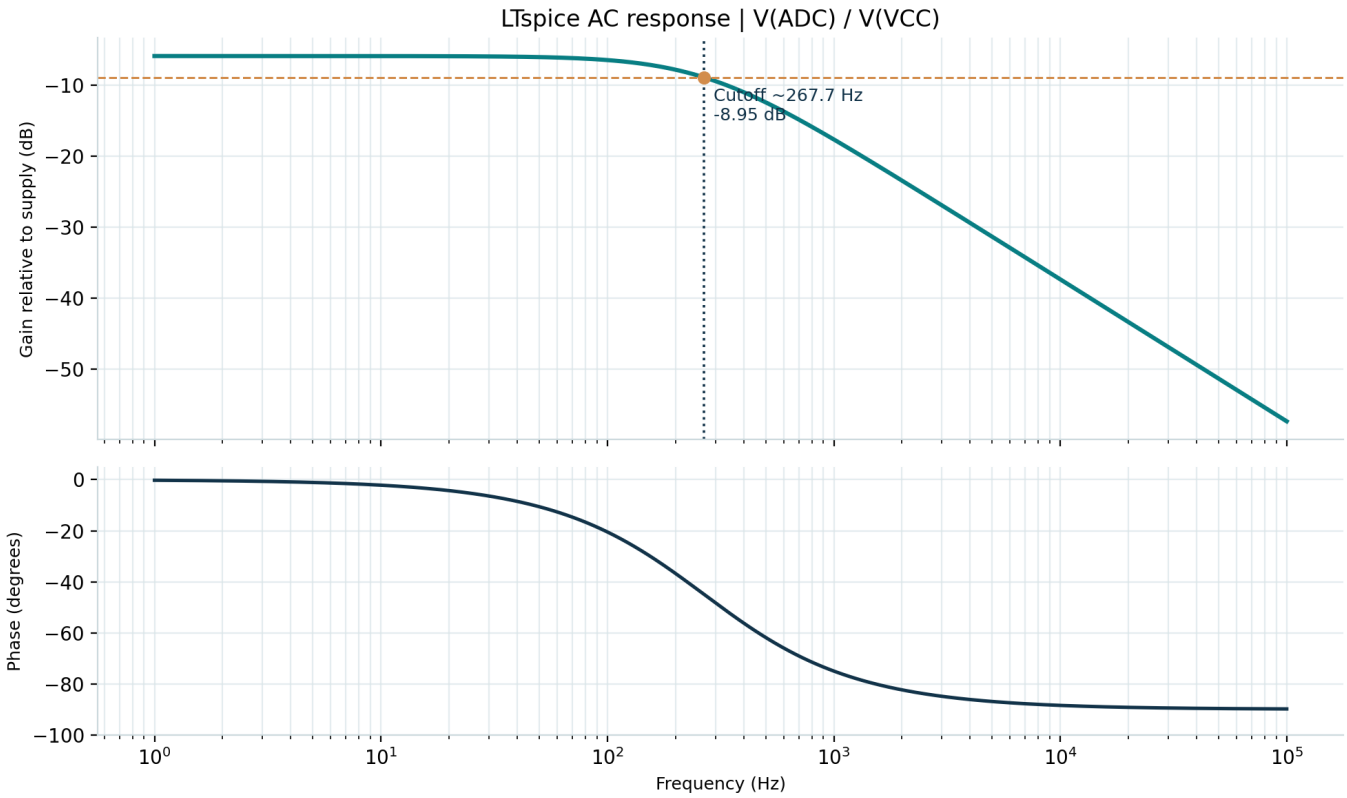
Independent temperature calibration remains pending. The uploaded LTspice netlist has a stale comment about five steps, but the saved directive and raw file actually contain only 25, 26 and 27°C.

14 / FREQUENCY RESPONSE

ADC Input Low-Pass Filter

LTspice AC analysis of R2 and C1, including divider source impedance

01 / ACTUAL SIMULATED RESPONSE



DC TRANSFER GAIN	EQUIVALENT R	-3 DB FREQUENCY
-5.94 dB	**5945 Ω**	**267.7 Hz**
9.82 kΩ / 10 kΩ divider	R2 + (R1 || RNTC)	Nominal circuit at 25°C

The correct LTspice trace is V(ADC)/V(VCC) (do not use a nested dB() call in the AC waveform viewer). Its high-frequency attenuation and phase shift are consistent with a first-order low-pass filter. Simulation does not establish real EMI performance or ADC noise reduction.

15 / HANDOFF

Simulation Evidence & Interview Notes

File map, limitations and a short explanation for technical interviews

01 / HOW TO REPRODUCE

TEMPERATURE

Open ThermoBench_LTspice.cir, run and plot V(adc). Open the SPICE Error Log for the three .meas voltages.

FREQUENCY

Open ThermoBench_Filter_AC.cir, run and add V(ADC)/V(VCC) to the AC waveform viewer.

SOURCE DATA

Read the uploaded .raw/.log files, or examine the 3-step and 501-point CSV exports supplied in the ZIP.

DOCUMENT

Use the LTspice PNG graphs along with the KiCad schematic, the PCB 3D images and hardware CSV plots.

02 / INTERVIEW-READY EXPLANATION

"I modeled my NTC voltage-divider and ADC filter in LTspice. I ran temperature-dependent simulations at 25, 26 and 27 degrees Celsius, and checked the predicted ADC voltages against an earlier multimeter observation. I also simulated the RC frequency response; its nominal cutoff was about 268 hertz. The circuit behavior is consistent with the model, but I still need an independent temperature reference to establish accuracy."

03 / REMAINING LIMITATIONS

PCB not fabricated or bench-verified. Reference thermometry and calibrated ADC comparison remain pending. The existing report records five ignored KiCad DRC check categories. Inferred temperature/model fit must not be stated as measured temperature accuracy.